

MINI PROJECT

Farm-to-Market Vegetable Price & Demand Analytics using Python, Power BI and Tableau

AIM

To analyze vegetable supply, demand, farmer prices, market prices, revenue and seasonal patterns using **Python, Power BI and Tableau**, and to create meaningful visualizations that help understand farm-to-market vegetable price and demand trends.

DOMAIN

Data Analytics and Visualization

Sub-Domain:

Agricultural and Market Analytics

PROBLEM STATEMENT

Vegetable prices and demand can change according to the vegetable type, season, market and quantity supplied. Farmers and market stakeholders need a simple way to understand these variations.

The project focuses on analyzing farm-to-market vegetable data to identify:

- Quantity of vegetables supplied
- Quantity of vegetables sold
- Vegetable demand
- Farmer price
- Market price
- Price difference between farmer and market
- Farmer revenue
- Market revenue
- Seasonal demand patterns
- Market-wise variations

A data analytics and visualization system is therefore required to convert the raw agricultural data into understandable charts and dashboards.

OBJECTIVES

The main objectives of the project are:

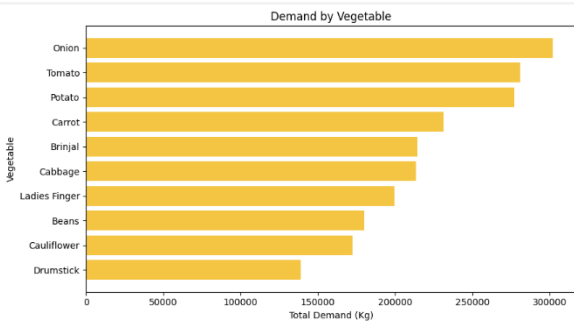
1. To collect and organize farm-to-market vegetable data.
2. To clean and prepare the dataset for analysis.
3. To analyze vegetable demand and supply.
4. To compare farmer prices with market prices.
5. To analyze monthly market price trends.
6. To study seasonal demand patterns.
7. To visualize the results using Python.
8. To create an interactive dashboard using Power BI.
9. To create visual analytics using Tableau.
10. To derive useful insights from the visualizations.

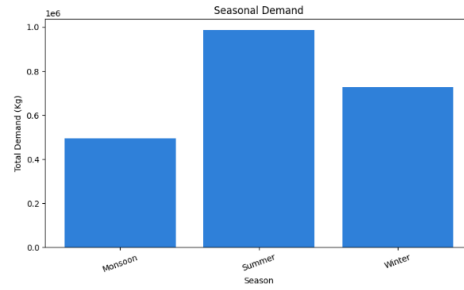
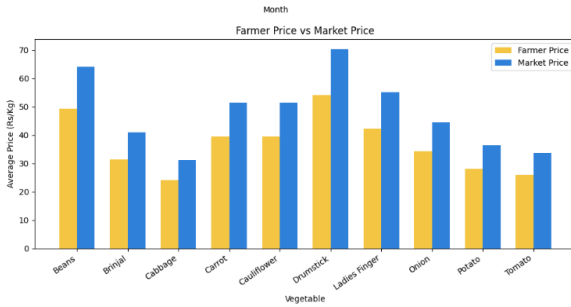
OUTPUT:

Python :

Dataset Shape: (5498, 29)

	Date	Vegetable	Farm	Market	Season	Quantity_Supplied_Kg	Quantity_Sold_Kg	Leftover_Kg	Farmer_Price_Rs_Per_Kg	Market_Price_Rs_Per_Kg	Demand_Kg	Farmer_Revenue_Rs	Market_Revenue_Rs	Price_Difference_Rs_Per_Kg	Price_Margin_Percent	Day	Month	Year	Market_Distance_Km
0	2025-01-01	Beans	Farm E	Avadi	Winter	340	92	248	40.07	52.90	105	3686.44	4830.00	12.43	31.02	Wednesday	January	2025	32
1	2025-01-01	Brinjal	Farm E	Madhavaram	Winter	755	263	492	32.65	45.75	244	8586.95	12032.25	13.10	40.12	Wednesday	January	2025	22
2	2025-01-01	Cabbage	Farm D	Ambattur	Winter	861	450	411	16.06	22.46	467	7227.00	10107.00	6.40	39.85	Wednesday	January	2025	20
3	2025-01-01	Carrot	Farm E	Tambaram	Winter	1076	282	794	42.25	55.57	301	11914.50	15670.74	13.32	31.53	Wednesday	January	2025	28
4	2025-01-01	Cauliflower	Farm D	Madhavaram	Winter	1025	321	704	41.66	55.07	328	13372.86	17677.47	13.41	32.19	Wednesday	January	2025	22





Total Quantity Supplied: 4088065 Kg
Total Quantity Sold: 2205842 Kg
Total Demand: 2210647 Kg
Average Farmer Price: 36.85 Rs/Kg
Average Market Price: 47.93 Rs/Kg

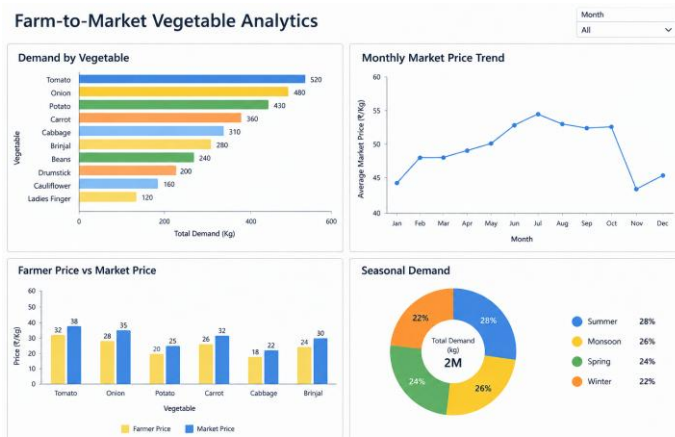
POWER BI :

Sales Performance Dashboard



TABLEU:

Farm-to-Market Vegetable Analytics



RESULT:

The **Farm-to-Market Vegetable Price & Demand Analytics** project was implemented using **Python, Power BI and Tableau**.

Python was used for data processing, aggregation and basic visualization. Power BI was used to create an interactive dashboard containing key performance indicators and analytical charts. Tableau was used to create interactive visualizations for exploring vegetable demand, prices and seasonal patterns.

The project demonstrates how data analytics and visualization techniques can be applied to agricultural market data to understand **supply, demand, prices, revenue and seasonal variations**.